

Problem Set – While Loops. Develop IPO for each of the problems below and then save within this document. Then Write the code files using C++. Upload both IPO and .CPP files to Blackboard.

1. Allow any number of users to enter a quantity and price at the keyboard (use ctrl+z to stop). Compute the extended price (quantity times price). If quantity is over 1000 give a 10% discount. Display quantity, price and extended, discount amount and discounted price for each entry. Keep a sum of the total for all the discounted prices. Display the total of discounted prices after all entries have been entered.

Input	Process	Output
• Quantity • Price Per	• Declare Extended Price, Discount Amount, Discounted Price • Declare Total Discounted Price and set at 0 • Assign a variable to the input for Quantity • Assign a variable to the input for Price Per • While not EOF • Calculate Extended Price by finding the product of Quantity and Price • If Quantity is greater than 1000, Calculate Discount Amount by finding the product Extended Price and .10 ○ Otherwise, set Discount Amount at 0 • Calculate Discounted Price by finding the difference of Extended Price and Discount Amount • Add Discounted Price to Total Discounted Price • Display Quantity, Price, Extended Price, Discount Amount, Discounted Price • Ask for new entry for Quantity and Price Per • End Loop • Display Total Discounted Price	<i>Each Entry</i> • Quantity • Price Per • Extended Price • Discount Amount • Discounted Price <i>After all Entries</i> • Total Discounted Price

2. Allow any number of players to enter last name, number of hits and at bats at the keyboard (use ctrl+z to stop). Compute batting average (hits/ at bats). Display last name and batting average for each player. Keep a count of the number of players (or entries) made. Display the count after all entries have been made.

Input	Process	Output
- Last Name - Number of Hits - Number of Bats	- Declare Batting Average - Declare Player Count and set at 0 - Assign a variable to the input for Last Name, Number of Hits, Number of Bats While not EOF - Calculate Batting Average by finding the quotient of Number of Hits and Number of Bats - Add 1 to Player Count - Display Last Name and Batting Average - Ask for new entry for Last Name, Number of Hits, Number of Bats End Loop - Display Player Count	<i>Each Entry</i> - Last Name - Batting Average <i>After all Entries</i> - Player Count

3. Enter destination city, miles travelled to get there and gallons of gasoline used for any number of trips entered at the keyboard (use ctrl+z to stop). Compute miles per gallon (miles travelled / gallons used). Display the destination city and miles per gallon for each trip entered. Sum the miles travelled and give a count of the number of trips made. Display these at the end of the program.

Input	Process	Output
- Destination City - Miles Travelled - Gallons Used	- Declare Trip-Mpg - Declare Total Miles and Number of Trips, set at 0 - Assign a variable to the input for Destination City, Miles Travelled, and Gallons Used While not EOF - Calculate Mpg by finding the quotient of Miles Travelled and Gallons Used - Add Miles Travelled to Total Miles - Add 1 to Number of Trips - Display Destination City and Trip-Mpg - Ask for new entry for Destination City, Miles Travelled, and Gallons Used	<i>Each Entry</i> - Destination City - Mpg <i>After All Entries</i> - Total Miles - Number of Trips

	• End Loop • Display Total Miles and Number of Trips	
--	--	--

4. Allow the employee to enter last name, job code and hours worked (use ctrl+z to stop). Calculate pay. (Job code L is \$25/hr, A is \$30/hr and J is \$50/hr). Give time and a half for overtime. Display last name, job code, hours worked and pay for each employee. Sum the pay for each employee as well as count the entries made. After all entries are made, compute and display the average pay and the number of entries made.

Input	Process	Output
• Employee's Last Name • Job Code • Hours Worked	• Declare Pay, Pay Rate, and Overtime • Declare Average Pay and Number of Employees, set at 0 • Assign a variable to the input for Employee's Last Name, Job Code, and Hours Worked • While not EOF • If Job Code equals L, set Pay Rate at \$25 ○ If Job Code equals A, set Pay Rate at \$30 ○ If Job Code equals J, set Pay Rate at \$50 ○ Otherwise, End Program • If Hours Worked is greater than 40, calculate Overtime by finding the difference of Hours and 40, then calculate Pay by finding the sum of 40 * Pay Rate and the product of Overtime and Pay Rate * 1.5 ○ Otherwise, calculate Pay by finding the product of Hours and Pay Rate • Add Pay to Average Pay • Add 1 to Number of Employees • Display Employee's Last Name, Job Code, Hours Worked, Pay • Ask for new entry for Employee's Last Name, Job Code, and Hours Worked • End Loop • Calculate Average Pay by finding the quotient of Average Pay and Number of Employees • Display Average Pay and Number of Employees	<i>Each Entry</i> • Employee's Last Name • Job Code • Hours Worked • Pay <i>After All Entries</i> • Average Pay • Number of Employees

5. Allow the user to enter student last name, credit hours and district code for any number of students (use ctrl+z to stop). Compute tuition owed. In district (code of I) is charged \$250 per credit hour. Out of district (code of O) is \$550 per credit hour. Display student name and tuition owed for each entry. Sum the amount of tuition owed for all students as well and the total credit hours taken and finally the number of students who entered data. Display total tuition, total credit hours taken and count of number of students at the end.

Input	Process	Output
• Student's Last Name • Credit Hours • District Code	• Declare Tuition, Credit Hour Rate • Declare Total Tuition, Total Credit Hours, and Number of Students, set at 0 • Assign a variable to the input for Student's Last Name, Credit Hours, and District Code • While not EOF • If District Code equals I (In-District), set Credit Hour Rate at \$250 ○ If District Code equals O (Out of District), set Credit Hour Rate at \$550 ○ Otherwise, End Program • Calculate Tuition by finding the product of Credit Hours and Credit Hours Rate • Add Tuition to Total Tuition • Add Credit Hours to Total Credit Hours • Add 1 to Number of Students • Display Student's Last Name and Tuition • Ask for new entry for Student's Last Name, Credit Hours, and District Code • End Loop • Display Total Tuition, Total Credit Hours, and Number of Students	<i>Each Entry</i> • Student's Last Name • Tuition <i>After All Entries</i> • Total Tuition • Total Credit Hours • Number of Students